

CelebA: non-IID

COMPLETED VALIDATION COHORT - 10 shared seeds (91001-91010)

Accepted snapshot: 700/700 records, 2026-09-28T10:20:44 UTC

Data distribution			non-IID ($\alpha = 5$)				
Category	Method	Metric	Benign	F-Flip	FedSA	S-DFA	Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	86.85 $\pm$ 5.45	89.51 $\pm$ 1.82	74.76 $\pm$ 13.38	74.51 $\pm$ 13.37	85.48 $\pm$ 2.58
		AEOD $\downarrow$	0.0428 $\pm$ 0.0137	0.0492 $\pm$ 0.0045	0.0644 $\pm$ 0.1097	0.0522 $\pm$ 0.0913	0.0267 $\pm$ 0.0157
		ASPD $\downarrow$	0.0958 $\pm$ 0.0353	0.1161 $\pm$ 0.0143	0.0614 $\pm$ 0.0893	0.0591 $\pm$ 0.0756	0.0741 $\pm$ 0.0247
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.59 $\pm$ 0.97	89.83 $\pm$ 0.92	71.45 $\pm$ 11.85	67.06 $\pm$ 11.91	86.18 $\pm$ 2.41
		AEOD $\downarrow$	0.0388 $\pm$ 0.0057	0.0462 $\pm$ 0.0081	0.0759 $\pm$ 0.0860	0.1307 $\pm$ 0.1184	0.0328 $\pm$ 0.0133
		ASPD $\downarrow$	0.1092 $\pm$ 0.0071	0.1163 $\pm$ 0.0071	0.0730 $\pm$ 0.0885	0.1171 $\pm$ 0.1067	0.0829 $\pm$ 0.0204
Robust FL	Median	ACC (%) $\uparrow$	86.22 $\pm$ 1.31	86.51 $\pm$ 1.24	84.47 $\pm$ 1.30	84.48 $\pm$ 1.35	85.47 $\pm$ 1.26
		AEOD $\downarrow$	0.0518 $\pm$ 0.0068	0.0594 $\pm$ 0.0084	0.0425 $\pm$ 0.0118	0.0461 $\pm$ 0.0142	0.0502 $\pm$ 0.0111
		ASPD $\downarrow$	0.0963 $\pm$ 0.0096	0.1067 $\pm$ 0.0097	0.0799 $\pm$ 0.0113	0.0839 $\pm$ 0.0124	0.0929 $\pm$ 0.0086
Robust FL	FLTrust	ACC (%) $\uparrow$	89.68 $\pm$ 0.60	89.83 $\pm$ 0.69	89.48 $\pm$ 0.70	89.48 $\pm$ 0.70	89.64 $\pm$ 0.71
		AEOD $\downarrow$	0.0445 $\pm$ 0.0049	0.0493 $\pm$ 0.0054	0.0441 $\pm$ 0.0063	0.0441 $\pm$ 0.0063	0.0475 $\pm$ 0.0052
		ASPD $\downarrow$	0.1132 $\pm$ 0.0060	0.1197 $\pm$ 0.0044	0.1120 $\pm$ 0.0041	0.1120 $\pm$ 0.0041	0.1153 $\pm$ 0.0053
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.39 $\pm$ 1.11	89.84 $\pm$ 0.76	59.95 $\pm$ 16.51	61.29 $\pm$ 15.59	78.37 $\pm$ 14.94
		AEOD $\downarrow$	0.0314 $\pm$ 0.0054	0.0394 $\pm$ 0.0103	0.0066 $\pm$ 0.0094	0.0108 $\pm$ 0.0161	0.0209 $\pm$ 0.0104
		ASPD $\downarrow$	0.1026 $\pm$ 0.0086	0.1109 $\pm$ 0.0090	0.0169 $\pm$ 0.0251	0.0206 $\pm$ 0.0317	0.0552 $\pm$ 0.0238
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.69 $\pm$ 1.68	90.07 $\pm$ 1.34	81.90 $\pm$ 11.66	80.44 $\pm$ 12.06	79.69 $\pm$ 13.94
		AEOD $\downarrow$	0.0357 $\pm$ 0.0055	0.0426 $\pm$ 0.0062	0.0400 $\pm$ 0.0214	0.0359 $\pm$ 0.0262	0.0408 $\pm$ 0.0124
		ASPD $\downarrow$	0.1085 $\pm$ 0.0102	0.1144 $\pm$ 0.0088	0.0851 $\pm$ 0.0350	0.0763 $\pm$ 0.0174	0.0892 $\pm$ 0.0276
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.59 $\pm$ 1.17	88.44 $\pm$ 0.74	88.41 $\pm$ 0.92	88.22 $\pm$ 0.78	88.34 $\pm$ 0.96
		AEOD $\downarrow$	0.0070 $\pm$ 0.0042	0.0100 $\pm$ 0.0070	0.0078 $\pm$ 0.0056	0.0126 $\pm$ 0.0093	0.0087 $\pm$ 0.0066
		ASPD $\downarrow$	0.0651 $\pm$ 0.0090	0.0589 $\pm$ 0.0092	0.0616 $\pm$ 0.0141	0.0584 $\pm$ 0.0189	0.0596 $\pm$ 0.0146

All 70 method/distribution/scenario cells use the same 10 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.

Round 70; validation only (19,867 images). Mean $\pm$ sample SD (ddof = 1); ACC in %, gaps on [0, 1].

AEOD denotes the implemented absolute TPR gap. No significance claim follows from mean ranks alone.

* Project adaptations. GuardFed includes training-root group calibration; baseline outputs are uncalibrated.

Recipes were selected on non-IID Benign/S-DFA, including seed 91001; this is coverage transfer, not IID-specific tuning.

This table uses 336 cu128 and 14 cu130 records; the source cohort contains 14 cu130 records. No 70-round equivalence claim.

Stage A covers seven implementations; ten further manuscript baselines and mechanism controls remain outstanding.